

《人工智能基础 A》 实验报告六

大模型微调

学号:_____ 姓名:_____

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学号

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实验报告

1. LLaMA-Factory 安装与部署

```
import pkg_resources
usage: python.exe -m llamafactory.cli [-h] [--model_name_or_path MODEL_NAME_OR_PATH]
[--adapter_name_or_path ADAPTER_NAME_OR_PATH] [--adapter_folder ADAPTER_FOLDER]
[--cache_dir CACHE_DIR] [--use_fast_tokenizer [USE_FAST_TOKENIZER]]
[--no_use_fast_tokenizer] [--resize_vocab [RESIZE_VOCAB]]
[--split_special_tokens [SPLIT_SPECIAL_TOKENS]] [--add_tokens ADD_TOKENS]
[--add_special_tokens ADD_SPECIAL_TOKENS]
[--new_special_tokens_config NEW_SPECIAL_TOKENS_CONFIG]
[--init_special_tokens {noise_init,desc_init,desc_init_w_noise}]
[--model_revision MODEL_REVISION] [--low_cpu_mem_usage [LOW_CPU_MEM_USAGE]]
[--no_low_cpu_mem_usage] [--rope_scaling {linear,dynamic,yarn,llama3}]
[--flash_attn {auto,disabled,sdpa,fa2,fa3}] [--shift_attn [SHIFT_ATTN]]
[--mixture_of_depths {convert,load}] [--use_unsloth [USE_UNSLOTH]]
[--use_unsloth_gc [USE_UNSLOTH_GC]]
[--enable_liger_kernel [ENABLE_LIGER_KERNEL]]
[--moe_aux_loss_coef MOE_AUX_LOSS_COEF]
[--disable_gradient_checkpointing [DISABLE_GRADIENT_CHECKPOINTING]]
[--use_reentrant_gc [USE_REENTRANT_GC]] [--no_use_reentrant_gc]
[--upcast_layernorm [UPCAST_LAYERNORM]]
[--upcast_lmhead_output [UPCAST_LMHEAD_OUTPUT]]
[--train_from_scratch [TRAIN_FROM_SCRATCH]]
[--infer_backend {huggingface,vllm,sglang}] [--offload_folder OFFLOAD_FOLDER]
[--use_kv_cache [USE_KV_CACHE]] [--no_use_kv_cache]
[--use_vl_kernels [USE_VL_KERNELS]]
[--infer_dtype {auto,float16,bfloat16,float32}] [--hf_hub_token HF_HUB_TOKEN]
[--ms_hub_token MS_HUB_TOKEN] [--om_hub_token OM_HUB_TOKEN]
[--print_param_status [PRINT_PARAM_STATUS]]
[--trust_remote_code [TRUST_REMOTE_CODE]]
[--quantization_method {bnb,gptq,awq,aqlm,quanto,eeq,hqq,mxfp4,fp8}]
[--quantization_bit QUANTIZATION_BIT] [--quantization_type {fp4,nf4}]
[--double_quantization [DOUBLE_QUANTIZATION]] [--no_double_quantization]
[--quantization_device_map {auto}] [--image_max_pixels IMAGE_MAX_PIXELS]
[--image_min_pixels IMAGE_MIN_PIXELS]
[--image_do_pan_and_scan [IMAGE_DO_PAN_AND_SCAN]]
[--crop_to_patches [CROP_TO_PATCHES]] [--video_max_pixels VIDEO_MAX_PIXELS]
[--video_min_pixels VIDEO_MIN_PIXELS] [--video_fps VIDEO_FPS]
[--video_maxlen VIDEO_MAXLEN] [--use_audio_in_video [USE_AUDIO_IN_VIDEO]]
[--audio_sampling_rate AUDIO_SAMPLING_RATE] [--export_dir EXPORT_DIR]
[--export_size EXPORT_SIZE] [--export_device {cpu,auto}]
[--export_quantization_bit EXPORT_QUANTIZATION_BIT]
[--export_quantization_dataset EXPORT_QUANTIZATION_DATASET]
[--export_quantization_nsamples EXPORT_QUANTIZATION_NSAMPLES]
[--export_quantization_maxlen EXPORT_QUANTIZATION_MAXLEN]
[--export_legacy_format [EXPORT_LEGACY_FORMAT]]
[--export_hub_model_id EXPORT_HUB_MODEL_ID] [--use_kt [USE_KT]]
```

2.

```
[--use_mca [USE_MCA]] [--use_hyper_parallel [USE_HYPER_PARALLEL]]
[--hyper_parallel_args HYPER_PARALLEL_ARGS] [--use_muon [USE_MUON]]
[--use_dft_loss [USE_DFT_LOSS]] [--use_asft_loss [USE_ASFT_LOSS]]
[--asft_alpha ASFT_ALPHA] [--use_eaft_loss [USE_EAFT_LOSS]]
[--eaft_alpha EAFT_ALPHA] [--freeze_vision_tower [FREEZE_VISION_TOWER]]
[--no_freeze_vision_tower]
[--freeze_multi_modal_projector [FREEZE_MULTI_MODAL_PROJECTOR]]
[--no_freeze_multi_modal_projector]
[--freeze_language_model [FREEZE_LANGUAGE_MODEL]]
[--compute_accuracy [COMPUTE_ACCURACY]]
[--disable_shuffling [DISABLE_SHUFFLING]]
[--early_stopping_steps EARLY_STOPPING_STEPS] [--plot_loss [PLOT_LOSS]]
[--include_effective_tokens_per_second [INCLUDE_EFFECTIVE_TOKENS_PER_SECOND]]
[--do_sample [DO_SAMPLE]] [--no_do_sample] [--temperature TEMPERATURE]
[--top_p TOP_P] [--top_k TOP_K] [--num_beams NUM_BEAMS]
[--max_length MAX_LENGTH] [--max_new_tokens MAX_NEW_TOKENS]
[--repetition_penalty REPETITION_PENALTY] [--length_penalty LENGTH_PENALTY]
[--skip_special_tokens [SKIP_SPECIAL_TOKENS]] [--no_skip_special_tokens]

options:
  -h, --help            show this help message and exit
  --model_name_or_path, --model-name-or-path MODEL_NAME_OR_PATH
                        Path to the model weight or identifier from huggingface.co/models or modelscope.cn/models.
                        (default: None)
  --adapter_name_or_path, --adapter-name-or-path ADAPTER_NAME_OR_PATH
                        Path to the adapter weight or identifier from huggingface.co/models. Use commas to separate
                        multiple adapters. (default: None)
  --adapter_folder, --adapter-folder ADAPTER_FOLDER
                        The folder containing the adapter weights to load. (default: None)
  --cache_dir, --cache-dir CACHE_DIR
                        Where to store the pre-trained models downloaded from huggingface.co or modelscope.cn.
                        (default: None)
```

数据准备与格式构建

2.1 数据集准备

```
1 {
2   "instruction": "请根据浙江大学校史，回答以下问题。",
3   "input": "浙江大学的校训是什么？有什么含义？",
4   "output": "浙江大学的校训是“求是创新”。 “求是”源于1938年竺可桢老校长提出的“求是”精神，意为追求真理、实事求是；“创新”则是后来为了适应时代发展而加入，鼓励师生勇于开拓、
5 },
6 {
7   "instruction": "请根据浙江大学校史，回答以下问题。",
8   "input": "浙江大学是什么时候建立的？",
9   "output": "浙江大学创建于1897年，前身为求是书院，是中国历史最悠久的高等学府之一。1928年改名为浙江大学，1952年院系调整后规模缩减，1998年原浙江大学、杭州大学、浙江农
10 },
11 {
12   "instruction": "请根据浙江大学校史，回答以下问题。",
13   "input": "竺可桢在浙江大学历史中有怎样的地位？",
14   "output": "竺可桢是浙江大学历史上最重要的校长之一，1936年至1949年间担任校长长达13年。他带领浙大西迁，在抗战烽火中坚持办学，将浙大发展成当时中国最具影响力的大学之一，
15 },
16 {
17   "instruction": "请根据浙江大学校史，回答以下问题。",
18   "input": "浙江大学西迁是什么历史事件？",
19   "output": "浙江大学西迁是抗日战争期间（1937-1946年）浙大为避免战火、坚持办学而进行的大规模迁徙。在竺可桢校长带领下，浙大辗转迁移，历经浙江建德、江西吉安、广西宜山，
20 },
21 {
22   "instruction": "请根据浙江大学校史，回答以下问题。",
23   "input": "浙江大学的校徽有什么含义？",
24   "output": "浙江大学的校徽以求是书院的古井为原型设计，整体呈圆形，中间是一口古井的俯视图，象征着浙大师生汲取知识、追求真理的精神。外环书写“浙江大学”和建校年份“1897”，
25 },
26 {
27   "instruction": "请根据浙江大学校史，回答以下问题。",
28   "input": "浙江大学的前身是什么？",
29   "output": "浙江大学的前身是求是书院，创建于1897年（清光绪二十三年），由杭州知府林启创办，是浙江省第一所新式高等学堂。1901年改名为浙江大学堂，1902年停办，1927年恢复
30 },
31 {
32   "instruction": "请根据浙江大学校史，回答以下问题。",
33   "input": "浙江大学的校训是什么？有什么含义？"
34 }
```

2.2 注册

数据集索引

```
### dataset
dataset: zju_history
template: deepseek
cutoff_len: 256
max_samples: 1000
preprocessing_num_workers: 2

### output
output_dir: saves/deepseek-r1-1.5b_zju_lora
logging_steps: 5
save_steps: 50
plot_loss: true
overwrite_output_dir: true

### train
per_device_train_batch_size: 1
gradient_accumulation_steps: 8
learning_rate: 1.0e-4
num_train_epochs: 3.0
lr_scheduler_type: cosine
warmup_ratio: 0.1
fp16: true

### eval (可选)
# val_size: 0.1
# per_device_eval_batch_size: 4
# evaluation_strategy: steps
# eval_steps: 50
```

```
columns: {  
  "prompt": "content"  
},  
  "folder": "python"  
}  
, },  
  "zju_history": {  
    "file_name": "zju_history.json",  
    "formatting": "alpaca"  
  }  
}
```

3. 微调参数设置与训练执行

3.1 复制模板

```
PS C:\Users\lenovo\LLaMA-Factory> cp examples\train_lora\qwen3_lora_sft.yaml config\deepseek_history_lora.yaml  
PS C:\Users\lenovo\LLaMA-Factory> |
```

3.2 修改关键参数

```
### model
model_name_or_path: C:/Users/lenovo/models/deepseek-1.5b
trust_remote_code: true

### method
stage: sft
do_train: true
finetuning_type: lora
lora_rank: 8
lora_target: all

### dataset
dataset: zju_history
template: deepseek
cutoff_len: 256
max_samples: 1000
preprocessing_num_workers: 1

### output
output_dir: saves/deepseek-r1-1.5b_zju_lora
logging_steps: 5
save_steps: 50
plot_loss: true
overwrite_output_dir: true

### train
per_device_train_batch_size: 1
gradient_accumulation_steps: 8
learning_rate: 1.0e-4
num_train_epochs: 3.0
lr_scheduler_type: cosine
warmup_ratio: 0.1
bf16: true
```

```
### model
model_name_or_path: Qwen/Qwen3-4B-Instruct-2507
trust_remote_code: true

### method
stage: sft
do_train: true
finetuning_type: lora
lora_rank: 8
lora_target: all

### dataset
dataset: identity,alpaca_en_demo
template: qwen3_nothink
cutoff_len: 2048
max_samples: 1000
preprocessing_num_workers: 16
dataloader_num_workers: 4

### output
output_dir: saves/qwen3-4b/lora/sft
logging_steps: 10
save_steps: 500
plot_loss: true
overwrite_output_dir: true
save_only_model: false
report_to: none # choices: [none, wandb, tensorboard, swanlab, mlflow]

### train
per_device_train_batch_size: 1
gradient_accumulation_steps: 8
learning_rate: 1.0e-4
num_train_epochs: 3.0
lr_scheduler_type: cosine
warmup_ratio: 0.1
bf16: true
ddp_timeout: 180000000
resume_from_checkpoint: null

### eval
# eval_dataset: alpaca_en_demo
# val_size: 0.1
# per_device_eval_batch_size: 1
# eval_strategy: steps
# eval_steps: 500
```

3.3 执行训练

```
(llama311) C:\Users\lenovo\LLaMA-Factory>
(llama311) C:\Users\lenovo\LLaMA-Factory>python -m llamafactory.cli train config\deepseek_history_lora.yaml
[transformers] warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps` instead.
[INFO|2026-06-03 06:44:09] llamafactory.hparams.parser:523 >> Process rank: 0, world size: 1, device: cuda:0, distributed training: False, compute dtype: torch.bfloat16
[INFO|configuration_utils.py:769] 2026-06-03 06:44:09,724 >> loading configuration file C:/Users/lenovo/models/deepseek-1.5b/config.json
[INFO|configuration_utils.py:847] 2026-06-03 06:44:09,739 >> Model config Qwen2Config {
  "architectures": [
    "Qwen2ForCausalLM"
  ],
  "attention_dropout": 0.0,
  "bos_token_id": 151643,
  "dtype": "bfloat16",
  "eos_token_id": 151643,
  "hidden_act": "silu",
  "hidden_size": 1536,
  "initializer_range": 0.02,
  "intermediate_size": 8960,
  "layer_types": [
    "full_attention",
    "full_attention",
    "full_attention",
  ],
}
[INFO|tokenization_utils_base.py:3278] 2026-06-03 06:44:36,365 >> chat template saved in saves/deepseek-r1-1.5b_zju_lora\chat_template.jinja
[INFO|tokenization_utils_base.py:2111] 2026-06-03 06:44:36,366 >> tokenizer config file saved in saves/deepseek-r1-1.5b_zju_lora\tokenizer_config.json
**** train metrics ****
epoch                =          3.0
total_flos           =       3332GF
train_loss           =        5.5346
train_runtime        =    0:00:11.88
train_samples_per_second =        2.524
train_steps_per_second  =        0.505
Figure saved at: saves/deepseek-r1-1.5b_zju_lora\training_loss.png
[WARNING|2026-06-03 06:44:36] llamafactory.extras.plotting:149 >> No metric eval_loss to plot.
[WARNING|2026-06-03 06:44:36] llamafactory.extras.plotting:149 >> No metric eval_accuracy to plot.
[INFO|modelcard.py:263] 2026-06-03 06:44:36,635 >> Dropping the following result as it does not have all the necessary fields:
{'task': {'name': 'Causal Language Modeling', 'type': 'text-generation'}}
(llama311) C:\Users\lenovo\LLaMA-Factory>
```

4.模型推理与效果验证

4.1 加载模型


```
### model
model_name_or_path: C:/Users/lenovo/models/deepseek-1.5b
# adapter_name_or_path: saves/deepseek-r1-1.5b_zju_lora
finetuning_type: lora
```

```
### dataset
template: deepseekr1
```

```
### generate
do_sample: false
max_new_tokens: 512
```